

Source-only model selection and limits of fixed spectral-sharing pipelines in cross-subject motor-imagery EEG decoding

Strict zero-calibration evaluation with separate external cohorts and BNCI sensor-level physiology

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7 October 2026

Abstract

Objective. We examined source-only training-duration selection and fixed shared mu/beta input pipelines for cross-subject motor-imagery EEG decoding. **Approach.** An exploratory four-class study used nine BNCI2014_001 participants, excluding both target sessions from each corresponding fit and selection. Separate frozen binary pipelines were evaluated on PhysioNet (109 people), Cho2017 (52) and Lee2019 offline-training runs (54, two sessions), without target fitting or adaptation. We re-audited saved predictions and source partitions. A downstream BNCI signal analysis, specified after decoder outcomes were known, described baseline-relative mu/beta power and associations with saved binary LOSO balanced accuracy. **Main results.** Mean-rank rather than mean-loss duration selection increased internal balanced accuracy from 33.72% to 42.67%; S3 and S8 contributed 87.57% of the aggregate gain. A matched-runtime fixed-duration control improved four of nine people despite a positive mean. Shared-minus-broad external differences were +0.573 percentage points (pp) on PhysioNet (95% person-bootstrap interval $[-0.097, 1.242]$), -1.513 pp on Cho2017 $[-2.249, -0.804]$ and $+0.204$ pp on Lee2019 $[-0.515, 0.969]$. Cho2017 and Lee2019 Holm-adjusted permutation p-values were 0.000100 and 0.604. These complete-pipeline contrasts used unequal source-selected durations. The BNCI signal analysis retained all 5,184 trials; its 2,592 hand-imagery trials gave mean signed laterality of -0.250 dB in mu and -0.168 dB in beta, with variable person-level signs. **Significance.** Training-duration choices yielded large but concentrated gains in this development benchmark. The shared-input pipelines offered no consistent external advantage. BNCI sensor patterns provide descriptive context; nine-person associations do not establish a decoder mechanism. Unknown voltage calibration and offline evaluation limit deployment conclusions.

Keywords: motor imagery; EEG; cross-subject decoding; EEGNet; source-only model selection; spectral parameter sharing; external evaluation; reproducibility.

1. Introduction

Motor-imagery EEG decoding must distinguish imagined actions in scalp recordings whose spatial and spectral structure varies between people. A calibration-free model should transfer to a new person without fitting to that person’s labels or distributional statistics. This differs from within-person evaluation and from adaptation using unlabeled target recordings. BCI Competition IV dataset 2a provides a small, well-documented benchmark [1,2], but its nine participants make a mean especially sensitive to a few large changes. Broader reviews and multi-dataset evaluation frameworks emphasize both variation between datasets and the dependence of reported performance on evaluation design [3,4].

A decoder that predicts predominantly one class can have balanced accuracy near chance even when the same architecture performs well with a different training duration. Such a failure can be mistaken for evidence that the input distribution requires normalization or a new representation. Conversely, a positive average after an intervention may reflect rescue of one or two participants rather than consistent improvement. Source-only fitting prevents per-fit target leakage, but it does not make a method independently confirmatory when the research program was developed after examining the same benchmark.

We separate three questions that are often merged in decoder comparisons. First, can heterogeneous source-validation loss curves lead to very short training schedules, and does within-fold rank aggregation change those schedules? Second, how does a pipeline processing predefined mu/beta views with one shared EEGNet compare with broad input when spatial, capacity, and robustness controls are reported? Third, how do those complete source-selected pipelines behave in external binary transfer? A separate descriptive component characterizes measured BNCI sensor signals without changing the decoders. The external evidence comprises an earlier PhysioNet evaluation [5,6] and a subsequently frozen, separately fitted pipeline for Cho2017 [7,8] and Lee2019 [9,10]. These are separate evaluations with different channel sets and time windows, not a pooled benchmark score.

This study examines training-duration selection and fixed spectral sharing under a source-only fitting constraint. We retain negative and adverse results, use participants rather than model seeds as the unit of inference, and distinguish exploratory development, runtime-matched controls and external evaluations. The design treats model selection as part of the domain-generalization method [11]. A separate post-outcome BNCI analysis provides descriptive signal context without further decoder fitting or inference.

2. Related Work

Spatial and spectral structure motivate classical motor-imagery decoders. CSP estimates discriminative spatial projections from class covariance structure [12], and filter-bank CSP combines spatial features across frequency bands [13,14]. These approaches motivate the spatial and spectral controls here, although the implemented filter bank is FBCSP-inspired rather than an exact reproduction. Its projections, feature scaling and classifiers are fitted on source partitions under the same target-information restriction as the neural models.

Compact convolutional models provide another representation family. EEGNet combines temporal filtering, depthwise spatial filtering and separable convolutions [15]; broader CNN work investigates learned EEG decoding and visualization [16]. Our shared-input model sends predefined mu and beta views through one configured EEGNet and averages class logits. Its parameter count matches one branch, while two-view computation and shared BatchNorm exposure differ. The tested bands are experimental views rather than verified physiological source separation.

Algorithm reviews and multi-dataset benchmarking emphasize variation between methods, people and datasets [3,4]. Model selection also belongs inside a domain-generalization evaluation [11]. We distinguish fitting restricted to source participants within a repeatedly used development benchmark from external evaluation of frozen pipelines. Methods that fit on target labels or unlabeled target distributions use a different information budget and were not compared directly in this study.

Recent domain-generalization studies address unseen-subject motor-imagery decoding through learned source-domain invariance. EEG-DG aligns marginal and conditional distributions across multiple source domains [17]. Zheng et al. combine spectral/task knowledge distillation with cross-source correlation alignment and distance regularization [18]. These studies motivate source-only generalization without target adaptation; their information budgets and selection procedures must still be checked for a matched comparison. The present study does not introduce a new domain-generalization algorithm or rank against those methods. It audits source-only model selection, participant heterogeneity and complete frozen-pipeline transfer, retaining adverse and null results with an inspectable evidence chain.

Recent EEG pretraining studies broaden the representation-learning landscape. LaBraM learns generic representations from large EEG collections [19], while EEGPT uses pretrained transformers for transferable EEG representations [20]. Their pretraining data and downstream fitting procedures introduce information and computational budgets that must be specified for a fair comparison. We did not evaluate either model or establish equivalence to our target-information contract; the present findings concern the selected source-only pipelines.

Event-related desynchronization and synchronization describe decreases and increases in spectral power relative to a reference interval [21]. Earlier motor-imagery work investigated sensorimotor rhythm changes [22]. These concepts motivate a separate measured-signal characterization, with the baseline and estimator stated explicitly. A scalp power pattern does not establish anatomical source localization or show that a learned decoder relied on that pattern.

3. Materials and Methods

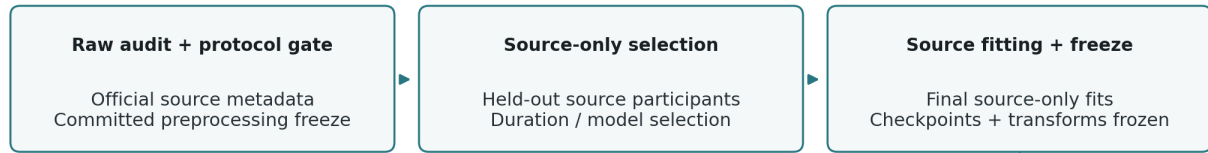
3.1. Experimental Design

The study separates exploratory development from two external-evaluation stages. The four-class BNCI2014_001 development stage holds out both sessions of the current target person from all corresponding fitting and duration selection. Earlier benchmark outcomes informed later candidates, so these nine-person comparisons remain exploratory. The earlier PhysioNet external evaluation (Q14) evaluates separately fitted binary BNCI models on PhysioNet; a metadata-driven sampling-rate amendment followed partial target execution. The frozen Cho2017/Lee2019 transfer experiment (Q15) fits a new binary source pipeline with a different montage, crop and reference, then evaluates its frozen broad, shared-input and CSP models on Cho2017 and Lee2019 offline-training segments. These datasets are external to fitting, but cross-provider identity nonoverlap was not independently established.

The primary external estimand is the equal-person mean paired balanced-accuracy difference between the frozen shared-input and broad pipelines, separately by cohort. The PhysioNet evaluation uses 17 shared and 18 broad epochs; the frozen Cho2017/Lee2019 evaluation uses 19 shared and 14 broad epochs, all selected on source data. The contrasts therefore evaluate complete selected pipelines rather than the isolated effect of sharing. CSP is contextual, and neither external stage tests rank versus mean-loss selection. The descriptive BNCI

physiological characterization (Q16) is a separate BNCI-only signal characterization defined after decoder results were known; its raw-signal summaries and associations cannot modify a decoder or establish its physiological mechanism.

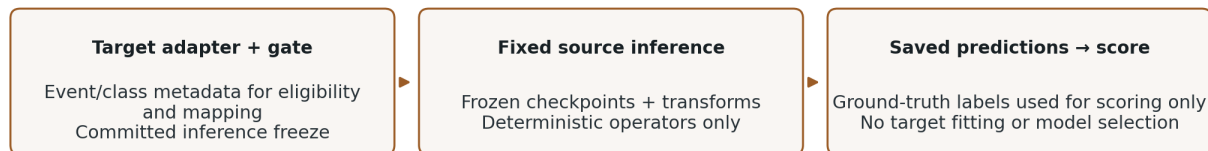
A Source-only learning



Q14 PhysioNet and Q15 Cho2017 / Lee2019 use separate source-trained artifacts.

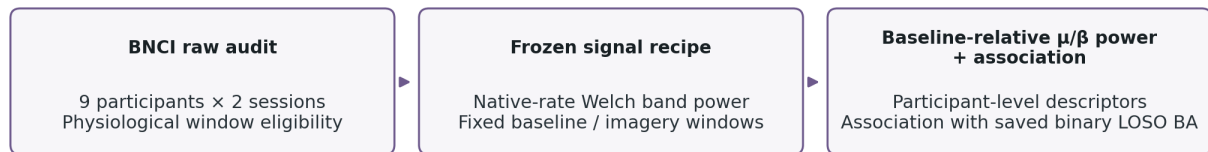
Frozen source artifacts

B Fixed target evaluation



No target-based model selection, optimization, normalization fit, or feedback to source learning.

C Separate descriptive physiology



No tuning of decoders; a descriptive association does not identify a learned physiological mechanism.

BA: balanced accuracy. Fixed CAR / resampling / channel maps are operations, not target-fitted parameters.

Figure 1. Study workflow and information boundaries. Both sessions of each internal target person are excluded from the corresponding source fit; the development programme remains exploratory. Q14 and Q15 use separate binary models and preprocessing contracts. Target inference applies deterministic adapters and frozen source-derived transformations with no fitting, adaptation or BatchNorm update. Event and class metadata support eligibility, canonical label mapping and scoring. A separate BNCI-only physiological branch reads native signals and labels after the decoder outcomes are known, without feedback into fitting or selection. People remain the aggregation unit. Q14's post-partial-run amendment and both external duration differences are detailed in the Methods and comparison-fairness table.

3.2. Datasets, Populations and Task Separation

BNCI2014_001 corresponds to BCI Competition IV dataset 2a [1,2]. Nine participants completed two recording sessions, each containing six runs of 48 trials: 12 each of left-hand, right-hand, feet, and tongue imagery. Twenty-two EEG channels and three EOG channels were sampled at 250 Hz. Original acquisition used a left-mastoid reference, right-mastoid ground, 0.5–100 Hz band-pass, and 50-Hz notch; the offline operations below are additional processing of those recordings. The primary neural input uses EEG channels only. The archived metadata contain 5,184 trials, 576 per participant and 1,296 per class. All 488 expert artifact-flagged trials remain in the primary evaluation population. This include-all endpoint differs from the original competition's artifact-free scoring; source-clean conditions exclude flagged source-training trials but retain every target trial.

The external EEG Motor Movement/Imagery Database version 1.0.0 [5] supplies a separate binary task. We include all 109 subjects and only imagery runs 4, 8, and 12; T1 and T2 denote left- and right-fist imagery. Executed-movement, bilateral-task, and rest events do not enter this contrast. The final audited inventory contains 327 official-checksum-verified EDF files

and 4,918 unique trials. The nominal 160-Hz acquisition description is supplemented by the actual headers: three subjects have 128-Hz recordings. Four-class and binary balanced accuracy have different chance levels and are never pooled into a common score.

Cho2017 [7,8] supplies left/right-hand imagery recordings from 52 people at a native 512 Hz. The audited evaluation retains 10,520 imagery trials: 49 people have 200 trials and three (S7, S9, S46) have 240. The Lee2019 motor-imagery data [9,10] contain 54 people recorded at 1,000 Hz in two sessions. We retain only EEG_MI_train, the provider’s offline-training runs, giving 100 trials per session and 10,800 trials in total. EEG_MI_test, other paradigms, and non-imagery events are outside this evaluation. The provider’s term “train” describes a recording segment: these external trials were used for inference and scoring, never for parameter fitting. We do not assert that the three external cohorts represent 215 independently verified unique identities across providers.

Table 1. Distinct populations and computational endpoints.

Analysis	People	Classes	Unique trials	Primary tensor
Internal BNCI	9	4	5,184	22 × 750 at 250 Hz
Q14 binary BNCI source	9	2	2,592	22 × 480 at 160 Hz
Q14 external PhysioNet	109	2	4,918	22 × 480 at 160 Hz
Q15 binary BNCI source	9	2	2,592	21 × 320 at 160 Hz
Q15 external Cho2017	52	2	10,520	21 × 320 at 160 Hz
Q15 external Lee2019	54	2	10,800	21 × 320 at 160 Hz

Each BNCI outer target has both sessions held out. Q14 and Q15 use separate binary source fits and preprocessing contracts. Neural inputs contain EEG channels only. Prediction-table rows repeat trials across models/seeds and are not independent participants; Lee sessions are pooled within person for the primary endpoint.

3.3. Strict Zero-calibration Information Contract

We define strict zero-calibration within each evaluation fit as inference with no parameter, state, preprocessing coefficient, hyperparameter, duration, seed, decision threshold or model choice fitted or selected using the current target person’s recordings or labels in that fit. Neural weights and BatchNorm statistics, shallow-model coefficients, source-fitted transforms and source-selected schedules are fixed before target inference. There is no target normalization fit, covariance alignment, adaptation, BatchNorm update, target-based stopping or seed selection. This restriction does not make the repeatedly used BNCI development programme independent of earlier target outcomes.

Target EEG supplies model inputs, and event and class metadata implement declared eligibility and canonical label mappings. Labels then score saved predictions. Cho recordings are stored in class-specific containers, so the adapter is not blind to native class metadata. Fixed filtering, channel selection, resampling and common-average referencing compute functions of target signals without estimating a target-dependent decoder. The Q15 Helmert basis depends only on channel count; scalars, CSP/PCA, whitening and EOG-regression coefficients, where used, are estimated on source-training data and applied unchanged. Separate physiological summaries may read raw signals and labels, but provide no feedback into bands, crops, weights, thresholds, schedules or model choice.

Zero target fitting is distinct from known physical acquisition calibration. Q15’s native-numeric-as-microvolt convention does not verify the exported voltage units. Earlier within-person and cross-session P2 models use target-person labeled calibration and remain outside the strict zero-calibration endpoints. The source-fitted P4 EOG regression uses

synchronous target EOG and therefore requires additional sensors even though it fits no target coefficients.

Table 2. Target information and fitting boundaries.

Operation	Information read	Target fitting	Interpretation
Eligibility, fixed channel and class maps, cue indexing	Native rate/channel/event/class metadata	None	Declared metadata adapter
Filtering, resampling, crop and Q15 CAR	Target EEG, fixed native metadata	None	Deterministic offline operations
Q15 Helmert coordinates	Channel count and signal	None	Analytical CAR-subspace basis
Source-fitted scaler/CSP/PCA/whitening/EOG regression	Source data to fit; target signal to apply	None	Frozen source coefficients; EOG needs extra sensors
Neural inference	Target tensors	None; weights and BatchNorm fixed	Strict zero-calibration prediction
Scoring and separate Q16 characterization	Predictions/raw signals and canonical labels	None	Downstream analyses with no decoder feedback
P2 within-person or cross-session models	Target-person labeled training trials	Yes	Calibrated context; separate endpoint

The restriction concerns fitting and selecting the corresponding decoder, rather than prohibiting all computations on target signals or all label access. Metadata inspection and downstream scoring/physiology do not make the development history outcome-blind.

3.4. Offline Preprocessing and Source-only Partitions

For internal BNCI analyses, fourth-order zero-phase Butterworth filters are applied within each native recording run before epoch extraction. The half-open interval [2.5,5.5) s after the MAT trial start corresponds to [0.5,3.5) s after the cue and yields 750 samples. There is no baseline subtraction, additional notch, or additional re-reference in the primary neural pipeline. MNE-loaded volt values are multiplied by 10^6 for network inputs in microvolts. These offline zero-phase operations use future samples and are not a causal online decoder.

The broad mean-loss and mean-rank EEGNet conditions use 4–40 Hz. Restricted broad input uses 8–30 Hz, and the shared representation uses fixed μ (8–13 Hz) and β (13–30 Hz) views. The bands are predefined experimental choices, not participant-specific estimates or physiological source separation. Any fitted scaler, CSP/PCA projection, or whitening transform is estimated on the applicable source-training partition only.

Each primary four-class outer split holds out both sessions of one participant. The eight source people contribute 4,608 trials and the target contributes 576. Sorted source IDs define four consecutive, two-person inner-validation groups. Each inner fit trains on six people/3,456 trials and validates on two people/1,152 trials. This separation is by person, not by random trial. Final evaluation retains three fixed seeds, averaged within person before cross-person aggregation. Model evaluation freezes BatchNorm statistics; target examples do not update them.

3.5. Training-duration Selection and Diagnostic Controls

The mean-loss rule selects the earliest epoch minimizing the equally weighted mean validation cross-entropy across the four inner folds. Candidate epochs are 1–40. The mean-rank rule first ranks epochs within each fold by validation cross-entropy, assigns average ranks to ties, averages those ranks across folds, and selects the earliest minimum. Its operational purpose is invariance to strictly increasing (order-preserving) transformations within a fold; that property does not guarantee better target prediction. The rank follow-up reuses the 36 frozen original inner trajectories and trains 27 new final models after the schedule has been frozen.

$$\text{Selected epoch} = \text{earliest argmin over } e \text{ of } (1/K) \sum_k \text{rank}_k[L_k(e)].$$

The rank candidate was motivated by earlier results on the same BNCI participants, so the internal comparison remains exploratory despite source-only per-fit selection. A four-cell diagnostic for S3 crosses raw versus source-normalized input with two versus 16 fixed epochs. Here raw means filtered, microvolt-scaled EEG without fitted source normalization. Two cells reuse the archived original conditions and two are new archived diagnostic conditions; this is one participant, not a population-level replication. Source normalization estimates per-channel mean and scale from source-training samples only.

Later controls compare fixed 20-epoch training with raw-loss-selected schedules. An additive matched-runtime experiment uses the same historical source-selected epoch counts but retrain the 27 final models on the same actual runtime and raw files as the fixed-duration comparator. It does not reselect epochs under the newer runtime. Source-count controls use $k = 2, 4$, or 6 source participants, four predefined identity subsets per count, and the $k = 8$ anchor; all counts use fixed 20 epochs. The 0train-versus-1test source-session contrast uses equal budgets: 2,304 trials, 20 epochs, and 720 optimizer updates per fit. Source-count contrasts and comparisons of one source session with the both-session anchor change the number of training examples and optimizer updates; those contrasts do not isolate a single causal factor.

The population raw-versus-source-normalized Q5/Q6 comparison also changes the selected duration for S3 from 2 to 16 epochs and S5 from 13 to 12. Its aggregate difference therefore does not isolate normalization. The fixed-duration Q7 four-cell diagnostic provides a more direct separation of normalization and duration for S3 only.

3.6. Neural Representation and Baseline Fairness

The four-class model is Braindecode 1.5.1 EEGNet with 22 input channels, 750 samples, $F1 = 8$, depth multiplier $D = 2$, $F2 = 16$, temporal kernel length 64 samples, and dropout 0.25. It has 2,932 trainable parameters in the archived implementation. Training uses Adam, learning rate 0.001, batch size 64, cross-entropy, zero weight decay, and no scheduler or default augmentation. The selection seed is 20260923 and the three final seeds are 20260924, 20260925, and 20260926. This is a configured EEGNet study, not an exact reproduction of the original paper’s preprocessing or splits [15].

For shared μ/β input, both band views pass through the same EEGNet in one concatenated mini-batch. Their class logits are averaged before cross-entropy in training and before softmax in inference. Network parameters and BatchNorm are shared; dropout acts per view. There are no learned band weights, attention mechanism, or separate independently trained branches. Parameter count matches one EEGNet, while forward-pass work and preprocessing differ.

$$\text{Shared prediction: } p = \text{softmax}[(f\theta(X\mu) + f\theta(X\beta))/2].$$

Controls include individual frequency bands; fixed source-selected durations; Welch log-bandpower with source-fitted shallow classifiers; eight source-only CSP or PCA time-series projections per band; log-Euclidean covariance features; source-only normalization; and source-clean training. The CSP/PCA neural controls both use 16 projected channels and source-fitted standardization, so their width is matched to each other but differs from the 22-channel shared-logit model. Architectural ablations use two independent networks, early band stacking, four shared bands, and a broadband capacity comparator. The independent two-band model has 5,864 parameters and its comparator has 5,914, a disclosed 0.85% mismatch. The traditional filter-bank implementation is FBCSP-inspired, not a literal reproduction of Ang et al. [13,14].

Table 3. Matching and remaining differences in decoder comparisons.

Comparison	Matched aspects	Remaining difference / scope
Q5 mean-loss vs Q8 mean-rank	Nine-person all-trial LOSO, 22×750, 4–40 Hz, original 36 inner trajectories, final seeds	Schedule changes; candidate arose after BNCI outcomes; exploratory
Q5 raw vs Q6 source normalization	Same all-trial targets and EEGNet family; source-only scaler fit	Selected epochs also change: S3 2→16, S5 13→12; not normalization alone
Q7 S3 raw/normalized ×2/16 epochs	Epochs fixed within each raw/normalized pair	One-person diagnostic
Q9 broad vs shared, reselected durations	Same targets, dimensions, EEGNet parameter count, optimizer/seeds	Bands, two-view computation, BatchNorm exposure and schedules differ
Q9-E002 fixed-schedule broad/shared	Same targets and frozen Q8 per-target epochs	Duration-controlled sensitivity; views/computation still differ
Q10 CSP vs PCA neural projections	Eight source-fitted components per band; 16 projected channels and source scaling	Matched to each other, not to 22-channel shared-logit input
Q11 independent branches vs capacity comparator	Same targets; approximate trainable capacity	5,864 vs 5,914 parameters; branch computation/normalization differ
Q12 GroupDRO vs balanced ERM	Same runtime and batching/sampling policy	Declared objective comparison
Q12 whitening/augmentation vs historical Q8	Same nominal target population and source-only rule	Historical baseline uses another runtime
Q13 fixed20 vs matched-runtime CE schedule	Same actual runtime, raw files and three seeds	Historical epochs replayed, not reselected in the newer runtime
Q13 source-session 0train vs 1test	2,304 source trials, 20 epochs, 720 updates	Session composition changes; both-session anchor also changes exposure
Q14 external broad vs shared	109 people/4,918 trials; 22×480; same source seeds and target rule	18 vs 17 epochs; metadata amendment after partial execution
Q15 external broad vs shared	Identical people/trials within cohort; 21×320 adapter, neural family/seeds	14 vs 19 epochs and one-view vs two-view processing
Q14/Q15 CSP4+LDA context	Same cohort scoring population; source-only fit	One shallow model vs three neural seeds; Q15 Helmet; differs from Q4 CSP8/OAS

These are implementation-specific controls rather than interchangeable model-class benchmarks. Matching selected aspects does not isolate every causal factor. Q14/Q15 external effects compare complete frozen pipelines; CSP comparisons remain descriptive.

3.7. Source-only Robustness Interventions

The six completed robustness conditions comprise source-pooled whitening, source-balanced empirical risk minimization (ERM), source-group distributionally robust optimization (GroupDRO), channel dropout, gain perturbation, and their combination. Source whitening uses equal-person covariance aggregation, a ridge of 5% of the mean eigenvalue, and scale preservation; no target covariance or target alignment is fitted. GroupDRO treats source person as group [23]: persistent log weights increase by 0.05 times detached per-group cross-entropy, are normalized by log-sum-exp, and weight the group loss. Its declared matched comparator is balanced ERM, preserving the batching/sampling intervention. Training-only channel dropout uses probability 0.1 with surviving-channel scale 1/0.9; gain perturbation uses $U[0.8, 1.2]$, applied before dropout in the combined condition. Validation and target inputs are not augmented.

Whitening and augmentation conditions are compared with the previously frozen broad rank-selected baseline. That historical baseline was executed on a different runtime from the robustness batch, so those cross-batch deltas combine intervention and runtime variation. GroupDRO versus balanced ERM is a cleaner within-runtime objective comparison. We retain this distinction beside the numerical table rather than attributing every delta to its named method.

3.8. Earlier Binary Source Freeze and PhysioNet Evaluation (Q14)

The binary BNCI source population contains 2,592 left/right trials, 288 per person. Filtering and epoching occur at 250 Hz before deterministic polyphase resampling by 16/25 to 160 Hz, giving 22×480 samples. The broad binary model uses 8–30 Hz; the shared model uses the fixed μ/β views. All-nine-source duration selection uses the validation groups people 1–2, 3–4, 5–6, and 7–9, with equal fold weighting. Frozen durations are 18 broad-model epochs and 17 shared-model epochs. Checkpoints, channels, labels, contrast, seeds, and preprocessing are frozen before external access. A source-only CSP4 + LDA comparator uses `reg = None` and `LDA solver = svd`; it does not inherit the internal CSP8 OAS/shrinkage-LDA configuration. The external neural contrast compares complete pipelines with unequal source-selected durations, rather than isolating the effect of parameter sharing.

A partial external run stopped at S88 when its 128-Hz header violated the original 160-Hz assumption. A versioned, metadata-driven amendment inspected remaining headers without using their predictions or scores. S88, S92, and S100 are filtered and epoched at their native rate, then resampled by 5/4 from 384 to 480 samples. The other 106 subjects retain native 160-Hz processing. The 87 completed earlier subject outputs are copied byte-for-byte and 22 additional subjects complete the cohort. No target statistics, fitted normalization, adaptation, seed selection, or class-performance-dependent branch is introduced. Because amendment followed partial external execution, the combined result is not a pristine execution of the original freeze.

The original aggregate validation failed when re-serialized CSV probabilities were subjected to an inappropriate exact floating-point equality check. A later cross-host runtime mismatch was a separate limitation and motivated a versioned portability amendment. Both events remain in the record. The versioned validation-only attempt verified official EDF checksums and event identities and recomputed saved probability/argmax consistency, subject metrics, and statistics. It passed with zero new fits and zero new model-inference rows. This audit must not be described as regenerating all external predictions from checkpoints on the later validation host.

3.9. Frozen Cho2017/Lee2019 Transfer (Q15)

The frozen Cho2017/Lee2019 experiment is a distinct binary transfer experiment, not a continuation of the Q14 checkpoints. All nine BNCI participants supply 2,592 left/right trials. The common ordered montage comprises Fz, FC3, FC1, FC2, FC4, C5, C3, C1, Cz, C2, C4, C6, CP3, CP1, CPz, CP2, CP4, P1, Pz, P2, and POz. Each trial is processed in a fixed native-rate context from -1.5 to $+4.5$ s relative to the cue. Common-average reference is computed across these 21 selected channels, followed by native-rate zero-phase second-order-section Butterworth filtering with prototype order four. Broad input uses 8–30 Hz; the shared views use 8–13 and 13–30 Hz. Deterministic polyphase resampling yields 160 Hz, followed by the half-open cue crop $[0.5, 2.5)$ s, equivalent to samples $[320, 640)$ of the resampled six-second context. The model tensor is therefore 21×320 , and the learned decoder cannot be compared to Q14 as if only the external provider changed.

The Q15 adapter treats native numeric values under a native-numeric-as-microvolt convention. It does not independently verify physical voltage calibration, the original Cho hardware reference, or hardware cue latency. The first 64 of the Cho payload’s 68 rows are EEG; four EMG rows are excluded. Channel mapping uses the publication’s numbered montage because the payload does not supply names. Cho native labels left = 1 and right = 2 are retained. Lee native labels left = 2 and right = 1 are deterministically mapped to canonical left = 1 and right = 2. Lee MATLAB cue indices are converted using $t - 1$; the provider’s stored segments align with Python t , leaving a disclosed one-sample (1-ms) indexing discrepancy that was not changed using outcome scores. The supplement records these adapter assumptions and file-level audit coverage.

Source-only duration selection uses four groups, people 1–2, 3–4, 5–6, and 7–9, candidate epochs 1–40, equally weighted mean within-fold ranks of cross-entropy, and the earliest minimum. Final durations are 14 broad epochs and 19 shared epochs. The selection seed is 20260923, and final seeds are 20260924–20260926. Training uses the same configured EEGNet family, optimizer, and fixed-band logit averaging described above, with the Q15 channel/time dimensions. A source-fitted CSP4 + LDA comparator uses a fixed 21-to-20 Helmert coordinate transform after common-average reference, followed by source-only fitting. The external neural comparison is between two complete frozen pipelines with different source-selected durations; it does not isolate parameter sharing or band decomposition alone. The completed source work comprises eight inner neural fits, six final neural fits, and one shallow fit: 15 in total. Migration reused those artifacts and added no source or target fit.

Official raw metadata for 18 BNCI files and all declared Cho/Lee files were audited before the preprocessing contract was committed; the pre-fit publication was recorded at 2026-10-04 01:33:03 UTC. Source completion was recorded at 01:52:42 UTC. The separate inference freeze was committed on 2026-10-05 at 09:31:05 UTC, before external prediction, and fixes channels, timing, label adapters, transformations, checkpoints, seeds, scoring units, and primary statistical procedures. Preprocessing may inspect event metadata to determine eligibility, but no external labels are used for fitting or model selection. BatchNorm remains in evaluation mode; there is no target alignment, normalization fit, adaptation, or post hoc seed selection. The repository-defined freezes follow external metadata inspection and are not independently registered preregistration.

All 52 Cho2017 and 54 Lee2019 participants were processed under the declared adapters. The archived independent computational validator replays raw-to-epoch and frozen-model prediction calculations, checks canonical events and scoring populations, verifies inference and source-artifact bindings, and recalculates aggregate statistics. Validation passed with the stated acquisition-calibration limitations. The supplementary reproducibility

documentation records the exact input, parameter, checkpoint and result revisions.

3.10. BNCI-only Physiological Signal Characterization (Q16)

The BNCI characterization performed a separate descriptive analysis of native BNCI2014_001 recordings: 18 files, 108 motor-imagery runs, nine people and both sessions, retaining all 5,184 four-class trials and their artifact flags in the primary population. The signals retain all 22 EEG sensors at native 250 Hz, with no extra bandpass, notch, resampling, reference transform, source scaler or learned projection. Three EOG channels are excluded. EEG column order follows the provider montage; individual MAT files do not contain channel-name strings. Native one-based trial starts are converted to zero-based samples; the cue is 500 samples after the trial start. The fixed cue-relative baseline is $[-1.5, -0.5]$ s and task window is $[0.5, 2.5]$ s. Before power calculation, the analysis requires rehashing all originals against the archived source provenance and checking labels, artifact flags, finite samples, the declared montage order and complete window bounds. All 18 source SHA-256 digests, 5,184 event identities and 228,096 trial-channel-band eligibility rows passed; no power-guard failure excluded a trial. The primary population retained all 488 flagged trials, including 246 of the 2,592 hand trials.

Welch power is defined separately within each trial's baseline and task windows using float64 native numeric values, a periodic Hann taper, 250-sample segments, 125-sample overlap, 250-point FFT, linear detrending, one-sided density and arithmetic segment averaging. This gives 1-Hz bins. Closed 8-13-Hz and 13-30-Hz masks are integrated with the trapezoidal rule in physical frequency; the 13-Hz boundary supplies half-bin weights to the adjacent integrals. The trial-level change is $10 \log_{10}(P_{\text{task}}/P_{\text{baseline}})$, in dB. Nonfinite or nonpositive powers require an explicit invalid-row receipt rather than an epsilon or clipped ratio. The 1-s baseline contributes one Welch segment and the 2-s task contributes three overlapping segments. Unequal log-power estimator variance can create a positive offset even without a true power change, so these dB summaries are not claimed to be unbiased ERD estimates.

Arithmetic mean trial dB is aggregated within person, session, class, channel and band, followed by equal averaging of the two sessions. This averages log ratios rather than taking the logarithm of averaged powers. Left/right-hand C3/C4 summaries use common trial eligibility. The signed hand descriptor is $L = [(C3_{\text{right}} - C4_{\text{right}}) + (C4_{\text{left}} - C3_{\text{left}})]/2$. A negative descriptor means the contralateral task/baseline change is lower than the ipsilateral change; it does not alone establish absolute contralateral ERD. C3, Cz and C4 absolute changes, separate hand terms, full scalp summaries, sessions and artifact strata are retained to show that distinction.

The parameter specification was frozen on 6 October 2026 after the decoder outcomes were known, before the new physiological calculation. The analysis is therefore descriptive rather than an independent confirmation. Associations use existing matched-binary Q14-E001 BNCI LOSO scores, averaged over the three fixed neural seeds within each of the same nine people; all-nine-source fits and external-cohort scores are ineligible. The broad EEGNet score is the designated descriptive endpoint. Six Spearman coefficients cover the three fixed models and two fixed bands, with no p-values, tertile groups or outcome-selected bands/models. Decoder imagery covers $[0.5, 3.5]$ s, whereas the physiological power analysis covers $[0.5, 2.5]$ s. No external physiology was calculated, and no physiological-analysis output fitted, selected or updated a decoder. The parameter freeze is commit 050e01b028aaab8e3d745934b13b2d17e9bb0a7a; its public readback preceded raw-power calculation. The preprocessing freeze, raw receipt, execution manifest and final independent validation bind the specification, inputs and completed tables in Q16-P001-BNCI-20261006.

3.11. Statistical Analysis

$$\text{Balanced accuracy} = (1/C) \sum_c TP_c / (TP_c + FN_c).$$

The held-out person is the inference unit. Seeds are paired and averaged within person; predefined source-identity subsets are averaged within person and source count. Shallow conditions have one prediction per trial rather than three independent neural seeds. Paired differences are expressed in balanced-accuracy percentage points (pp). Person-bootstrap percentile intervals resample people, not trials, sessions, seeds, or source subsets. External intervals condition on the frozen source models and do not quantify source-cohort sampling uncertainty. Internal analyses remain exploratory because benchmark outcomes informed the research program and outer training sets overlap. For Lee2019, the two retained sessions are pooled before calculating each person/seed balanced accuracy. We then average the three neural-seed scores within person; this is an average of scores, not an ensemble of probabilities. A single CSP score is retained per person. Cross-cohort differences in scores are descriptive because Q14 and Q15 differ in source preprocessing and checkpoints.

We preserve archived experiment-specific statistics and identify each interval’s resampling procedure. Rank-selection intervals use 20,000 person-bootstrap draws with seed 20260923. The archived shared-minus-broad representation contrast and robustness intervals use 20,000 draws with seed 20260924. Source-composition and matched-runtime intervals use 10,000 draws with seed 20260926 and exhaustive two-sided sign flips over nine participant differences; familywise exploratory p-values use Holm correction within the frozen families. The earlier PhysioNet primary contrast uses 20,000 draws with seed 20260924 and an exact two-sided binomial sign test excluding exact zero differences. Its primary hypothesis is shared-minus-broad; contextual comparisons with CSP do not replace it. A supplementary numerical audit uses 200,000 draws with seed 20261002 and is labelled separately. We do not equate an interval excluding zero with agreement across all small-sample tests. For Q15, the predeclared family contains exactly two shared-minus-broad contrasts, one for each external cohort. Percentile person-bootstrap intervals use 20,000 draws with seed 20260924 [24]. Two-sided person-level sign-flip tests use 20,000 Monte Carlo draws with seed 20261003, assuming sign exchangeability of paired person effects under the null, evaluating the absolute mean, and applying the $(\text{exceedances} + 1) / (20,000 + 1)$ correction [25]. Holm adjustment controls this two-contrast family [26]; it does not retroactively incorporate the earlier PhysioNet or internal hypotheses. The smallest attainable Monte Carlo p-value is $1/20,001$. The intervals are unadjusted 95% intervals, and CSP comparisons are descriptive context. Descriptive Q15 counts of positive, negative, and tied person effects use an absolute 10^{-15} tolerance for ties; the frozen primary bootstrap and sign-flip calculations retain the saved numerical effects.

One historical rank-selection script labelled a nine-person binomial calculation as a sign test while retaining two unchanged participants in its denominator. We describe its $p = 0.1797$ as an archived nine-person sign sensitivity. A conventional tie-excluding calculation uses seven nonzero changes and gives $p = 0.015625$; the paired t-test gives $p = 0.111$. This distinction is reported transparently rather than selecting the favorable procedure or rewriting the frozen result. Multiplicity corrections from later declared families are not retrospectively imposed on unrelated earlier statistics.

The physiological characterization is a post-decoder-outcome descriptive component. Its physiological aggregation and decoder association use nine person-level records, without treating channels, trials, sessions or seeds as independent people. The six Spearman coefficients are descriptive; no physiological p-values or confirmatory family are introduced. They are separate from Q15’s frozen two-cohort tests and cannot retrospectively confirm the decoder contrasts.

3.12. Reproducibility and Validation Scope

Saved predictions were independently reaggregated to check trial identities, scoring populations, person-level confusion matrices, seed-averaged scores and reported statistical results. This numerical reanalysis did not load decoder checkpoints or generate predictions. The Cho2017/Lee2019 experiment additionally has a completed raw-to-epoch and frozen-model prediction replay; the earlier PhysioNet portability audit rechecked saved outputs without replaying inference. Validation coverage for earlier experiments remains experiment-specific and is documented in the supplement.

The physiological analysis was checked independently against native BNCI recordings. An explicit detrending, periodic-Hann FFT and trapezoidal integration calculation reproduced all 31,104 C3/Cz/C4 trial-band pairs: the maximum relative power difference was 1.85×10^{-15} and the maximum absolute dB difference was 7.11×10^{-15} , within fixed 10^{-10} tolerances. Raw spectral replay covered these three central channels; identities and eligibility were checked for all 228,096 saved rows, and all aggregation and association tables were recomputed. Numerical agreement establishes implementation consistency, not independent laboratory replication.

OpenAI Codex assisted analysis-code development and verification-code preparation. The computations use explicitly specified algorithms; the archived validation reports document the numerical checks. AI-assisted code generation is distinct from a learned decoder or a physiological model.

4. Results

4.1. Traditional baselines and heterogeneous duration selection

The all-trial four-class broad CSP8 + LDA baseline achieved 39.04% balanced accuracy; broad CSP + SVM and the full filter-bank LDA comparator achieved 37.50% and 38.12%. Raw mean-loss-selected EEGNet achieved 33.72%, below the broad CSP mean. These descriptive comparisons do not establish superiority of a model class. The mean-loss schedule selected only one or two epochs for S2, S3, and S8, while source-validation optima differed substantially across folds.

Source-only per-channel normalization increased the mean to 37.93%, a +4.21 pp difference. The archived paired 95% interval spans -1.12 to $+13.49$ pp and the median change is only +0.23 pp. The intervention has heterogeneous effects, including negative transfer in S8. It does not establish a general normalization benefit. Its selected schedules also changed for S3 (2 to 16 epochs) and S5 (13 to 12), so this population contrast combines normalization and duration changes.

The S3 diagnostic separates normalization from duration. Raw input at two epochs gave 26.10% balanced accuracy and source normalization at two epochs gave 25.69%. At 16 epochs, raw and normalized inputs gave 67.71% and 65.86%. The short-run raw condition placed approximately 94% of predictions in one class. Duration can reproduce the observed rescue for this participant; this four-cell diagnostic does not prove that duration explains all cross-person failures.

Mean-rank selection increased equal-person mean balanced accuracy to 42.67%, a +8.95 pp change over raw mean-loss selection. The archived bootstrap interval is $[+1.05, +19.96]$ pp and the median change is +1.85 pp. Seven participants improved and two were unchanged. S3 and S8 account for 87.57% of the summed improvement. Thus the positive average includes a marked concentration of benefit, and the prior same-dataset development and disagreement among small-sample statistics limit a general claim.

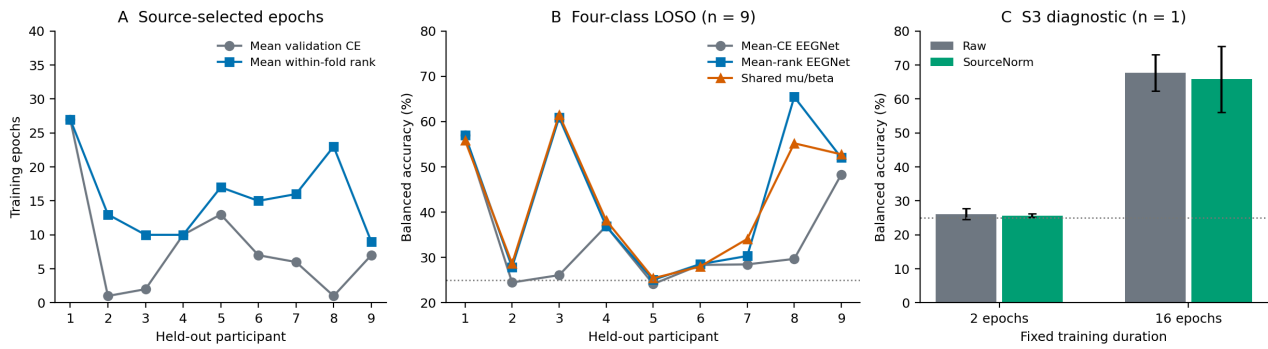


Figure 2. Training-duration selection and participant-level internal results. A: archived source-selected epochs. B: each participant’s all-trial balanced accuracy after averaging three fixed seeds; the dotted line is four-class chance (25%). C: the S3 four-cell diagnostic; error bars are standard deviations across three seeds, not uncertainty across people. Raw and source-normalized input are compared at two fixed durations. The diagnostic involves one person.

Table 4. Selected completed four-class conditions; all nine participants.

Condition	Bands (Hz)	Mean BA (%)	Interpretation
Broad CSP8 + LDA	4-40	39.04	Single source fit/fold
Filter-bank CSP + LDA	4-40 bank	38.12	Fixed filter-bank control
EEGNet, mean loss	4-40	33.72	Source-selected duration
EEGNet, source norm	4-40	37.93	Source-only fitted scale
EEGNet, mean rank	4-40	42.67	Reused source curves
Shared mu/beta	8-13 / 13-30	42.19	Representation-specific rank
Broad, fixed 20	4-40	43.99	Later runtime
Broad, matched CE schedule	4-40	33.92	Same runtime as fixed 20

All-trial primary population. Neural means average three fixed seeds within person first. Rows span different batches; only explicitly matched comparisons isolate runtime variation. These rows summarize declared controls, not a target-selected ranking. Complete representation and robustness inventories are supplied in the supplement.

4.2. Matched-runtime duration and source-composition controls

In the additive matched-runtime experiment, the historical mean-loss duration schedule gave 33.92% balanced accuracy, versus 43.99% for fixed 20 epochs. The fixed-minus-historical-schedule mean was +10.06 pp, with bootstrap 95% interval [+0.68,+21.53] pp and exact sign-flip $p = 0.125$. Only four of nine people improved, five worsened, and the median difference was -0.174 pp. The positive mean therefore reflects large recoveries rather than a typical-person benefit. Matching runtime retains the average schedule contrast, but does not establish 20 epochs as universally preferable; historical epoch counts were reused rather than newly selected under that runtime.

Mean balanced accuracy increased descriptively with source count. Broad fixed-duration models gave 32.96%, 38.89%, 43.16%, and 43.99% at $k = 2, 4, 6$, and 8 ; shared models gave 34.74%, 38.89%, 42.48%, and 43.98%. Source-identity variation within each count remained substantial. The two-source broad-minus-eight-source difference was -11.03 pp, but its Holm-adjusted exploratory p was 0.164. Every source-count/selection/session family retained

null corrected findings. One-session-versus-the-other contrasts were small and heterogeneous. These trajectories are consistent with training-information sensitivity and do not separate source diversity from sample count or optimizer-update budget.

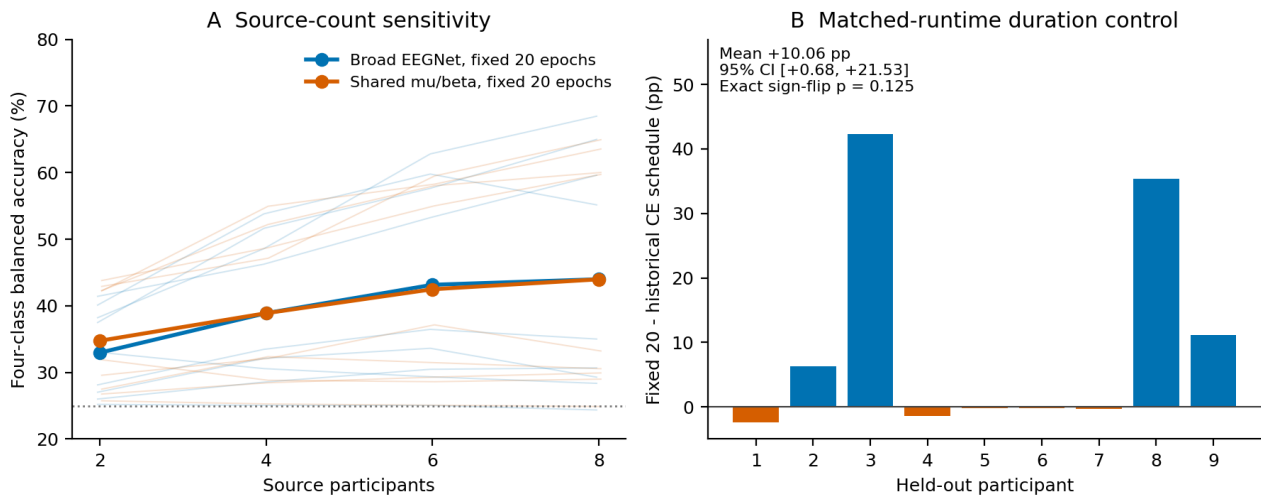


Figure 3. Source composition and matched-runtime controls. A: strong lines are equal-person means; faint lines are individual target-person trajectories after averaging the four predefined identity subsets and three seeds at $k = 2, 4, 6$. All counts use 20 epochs; $k = 8$ reuses the fixed-20 anchor. B: all nine fixed-20-minus-historical-CE-schedule effects on the matched runtime. The interval is the sign-reversed archived matched contrast. Source count changes data volume and optimizer updates; these are exploratory sensitivities.

4.3. Spectral, spatial, and architectural interventions

The internal shared mu/beta condition achieved 42.19% versus 42.67% for the broad mean-rank baseline, a -0.48 pp mean difference. The archived Q10-V001 exploratory paired interval is $[-3.21, +1.41]$ pp (20,000 draws, seed 20260924), with exact sign-flip $p = 0.914$. There is no demonstrated primary spectral-sharing gain. A restricted broad band and the individual bands provide descriptive frequency controls, while fixed-duration representation controls preserve their declared schedules.

The source-only CSP8 and PCA8 neural projections achieved 41.48% and 42.06%. The architectural controls achieved 42.99% for independent two-band networks, 42.19% for the broadband capacity comparator, 42.66% for early stacking, and 39.80% for four-band sharing. These completed alternatives do not supply evidence for a general shared-band advantage. Their differences in preprocessing, projected width, BatchNorm behavior, and compute prevent a single-factor physiological interpretation. Complete model means and participant-level values accompany the manuscript.

4.4. Source-only robustness has no corrected established gain

Source-pooled whitening, source-balanced ERM, and gain perturbation had positive point estimates relative to the historical broad rank baseline, whereas dropout conditions were lower. None established a corrected improvement. The matched GroupDRO-minus-balanced-ERM difference was -6.45 pp (unadjusted bootstrap interval $[-11.47, -1.79]$ pp; exact sign-flip $p = 0.046875$, Holm-adjusted $p = 0.140625$). This adverse estimate is a follow-up lead, not evidence that distributionally robust optimization generally harms EEG decoding. Whitening and gain perturbation intervals cross zero; their cross-runtime historical comparator further limits attribution.

Table 5. Exploratory source-only robustness (nine participants).

Condition	BA (%)	Control	Δ (pp)	95% CI (pp)	Holm p
Whitening	44.34	Q8 historical	+1.67	[−0.39, +3.67]	0.3594
Balanced ERM	43.18	Q8 historical	+0.51	[−0.71, +1.77]	0.4766
GroupDRO	36.73	Balanced ERM	−6.45	[−11.47, −1.79]	0.1406
Channel dropout	41.00	Q8 historical	−1.67	[−4.51, +0.64]	0.7617
Gain perturbation	43.42	Q8 historical	+0.75	[−0.34, +1.86]	0.7617
Dropout + gain	40.92	Q8 historical	−1.75	[−5.05, +0.91]	0.7617

Intervals are the archived, unadjusted person-bootstrap intervals; p-values use the declared within-family Holm correction. All comparisons with Q8 cross runtimes. GroupDRO versus balanced ERM is the matched-batching, within-runtime contrast. No interval/p-value is treated as a target-selection rule.

4.5. Earlier PhysioNet transfer: a small, uncertain primary effect

In binary BNCI development, broad, shared, and CSP models achieved 68.21%, 67.35%, and 61.54% balanced accuracy. These development outcomes cannot be pooled with the four-class analyses. The separately frozen all-nine-source models then provided predictions for all 109 external subjects. The final validation confirms 4,918 trial identities and 34,426 saved prediction rows. No target fit was made.

External equal-person balanced accuracy was 61.81% for broad EEGNet, 62.39% for shared mu/beta, and 54.46% for CSP4 + LDA. The predeclared shared-minus-broad primary difference was +0.573 pp, with 95% bootstrap interval [−0.097, +1.242] pp. Sixty-three people favored shared, 43 favored broad, and three tied; the exact tie-excluding sign-test p was 0.06446. Neither the interval nor this test establishes external superiority.

Individual performance varied widely. The shared model ranged from 39.33% to 97.07% balanced accuracy, with median 58.73%. Three-seed average differences also varied in direction across people. The three resampled subjects had mean shared-minus-broad Δ = +0.514 pp, versus +0.575 pp for the 106 native-160-Hz subjects. The small three-person subgroup is descriptive, not a powered subgroup test. This result is reported with the sampling-rate and validation-portability amendments, rather than as an unmodified original external protocol.

Table 6. Separate binary external evaluation (109 participants).

Frozen model	Mean BA (%)	Subject SD (pp)	Median BA (%)
Broad EEGNet	61.81	12.56	58.23
Shared mu/beta	62.39	12.92	58.73
CSP4 + LDA	54.46	7.90	52.08

Deep-model seeds are averaged within each person. The primary contrast is shared minus broad: +0.573 pp, 95% CI [−0.097, +1.242], p = 0.06446. CSP comparisons are contextual and do not replace the primary hypothesis. Sampling-rate amendment and versioned portability validation are disclosed in Methods.

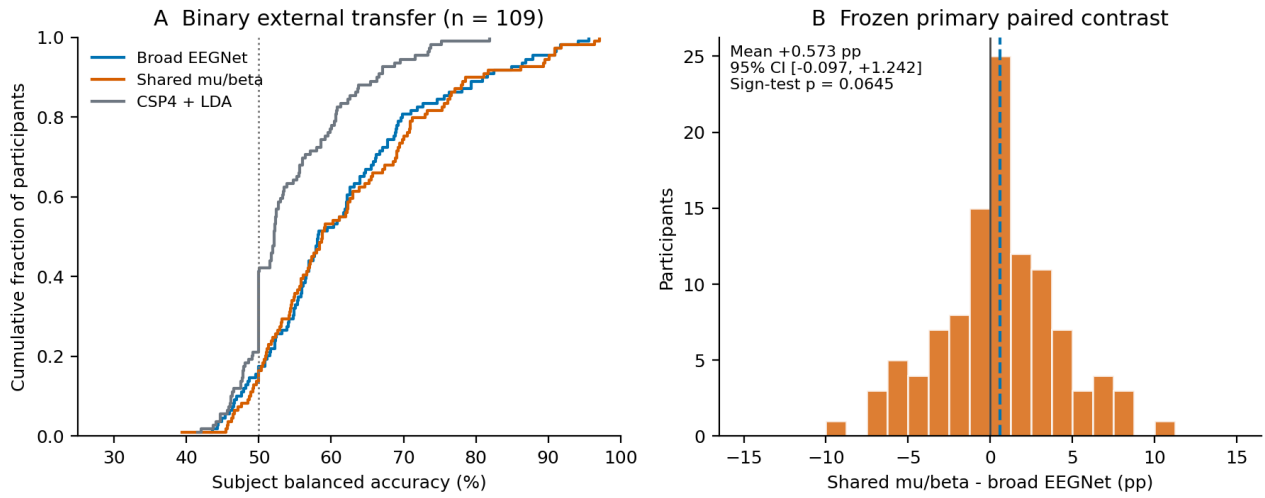


Figure 4. Separate-cohort binary transfer. A: empirical cumulative distributions of three-seed subject mean balanced accuracy (one CSP value per person); the vertical dotted line denotes binary chance (50%). B: all 109 primary paired effects, with zero and the mean indicated. The interval and p-value are the frozen person-bootstrap and tie-excluding sign-test outputs. The cohort contains three deterministically resampled subjects under the disclosed metadata amendment.

4.6. Completed Cho2017 and Lee2019 transfer: lower balanced accuracy on Cho2017 and an uncertain Lee2019 contrast

The completed Q15 outputs cover 21,320 unique external trials and 149,240 saved model/seed prediction rows, seven per trial. All canonical labels and event identities, person-level confusion matrices, aggregate balanced accuracies, frozen statistical outputs, and declared artifact bindings passed independent checks. Original source fits remained 15, with zero new migration fits and zero target fits. These counts demonstrate adherence to the source-only fitting rule; they are not independent repetitions of the external experiment.

On Cho2017, equal-person balanced accuracy was 59.82% for broad EEGNet and 58.31% for shared mu/beta EEGNet. The primary shared-minus-broad mean was -1.513 pp (95% person-bootstrap interval $[-2.249, -0.804]$ pp), with two-sided Monte Carlo sign-flip $p = 0.000049998$ and two-cohort Holm-adjusted $p = 0.000099995$. The raw p-value is the Monte Carlo procedure's minimum attainable value, rather than an exact probability below that resolution. Twelve people favored shared, 37 favored broad, and three were tied within a 10^{-15} numerical tolerance. This is a corrected adverse result for the declared complete-pipeline contrast.

On Lee2019, broad and shared models achieved 65.51% and 65.72% balanced accuracy. The primary difference was $+0.204$ pp (95% interval $[-0.515, +0.969]$ pp); raw and Holm-adjusted p-values were both 0.60367. Twenty-three people favored shared, 30 favored broad, and one tied. This result provides no evidence of a reliable improvement and is not an equivalence test. The two sessions were combined within each person, so neither the 108 session records nor the repeated model seeds increase the primary sample size beyond 54.

Table 7. New external binary evaluation under the Q15 freeze.

Cohort / model	BA (%)	Person SD (pp)	Left recall (%)	Right recall (%)
Cho2017 / Broad EEGNet	59.82	8.64	46.79	72.86
Cho2017 / Shared mu/beta	58.31	7.22	37.52	79.10
Cho2017 / CSP4 + LDA	51.99	5.67	94.10	9.88
Lee2019 / Broad EEGNet	65.51	10.97	75.56	55.46
Lee2019 / Shared mu/beta	65.72	10.76	72.69	58.75
Lee2019 / CSP4 + LDA	52.77	5.57	85.98	19.56

Neural scores and class recalls are averaged over three seeds within person, then equally over people. Person SD describes heterogeneity, not a standard error. Lee uses only offline-training runs, pooled across two sessions. Primary shared-minus-broad contrasts are -1.513 pp [$-2.249, -0.804$], Holm $p = 0.000100$ for Cho, and $+0.204$ pp [$-0.515, +0.969$], Holm $p = 0.604$ for Lee. Class recalls and CSP are descriptive; no additional confirmatory tests were added.

The CSP comparator was near binary chance on both cohorts and strongly favored left-hand labels. On Cho2017, the shared-input pipeline had lower mean left-hand recall from 46.79% to 37.52%, while right-hand recall increased from 72.86% to 79.10%. On Lee2019, the corresponding recalls changed from 75.56% to 72.69% and from 55.46% to 58.75%. These asymmetric changes are retained rather than hidden by balanced accuracy. They were calculated from the frozen outputs as descriptive checks and did not trigger any threshold, label, adapter, or model change.

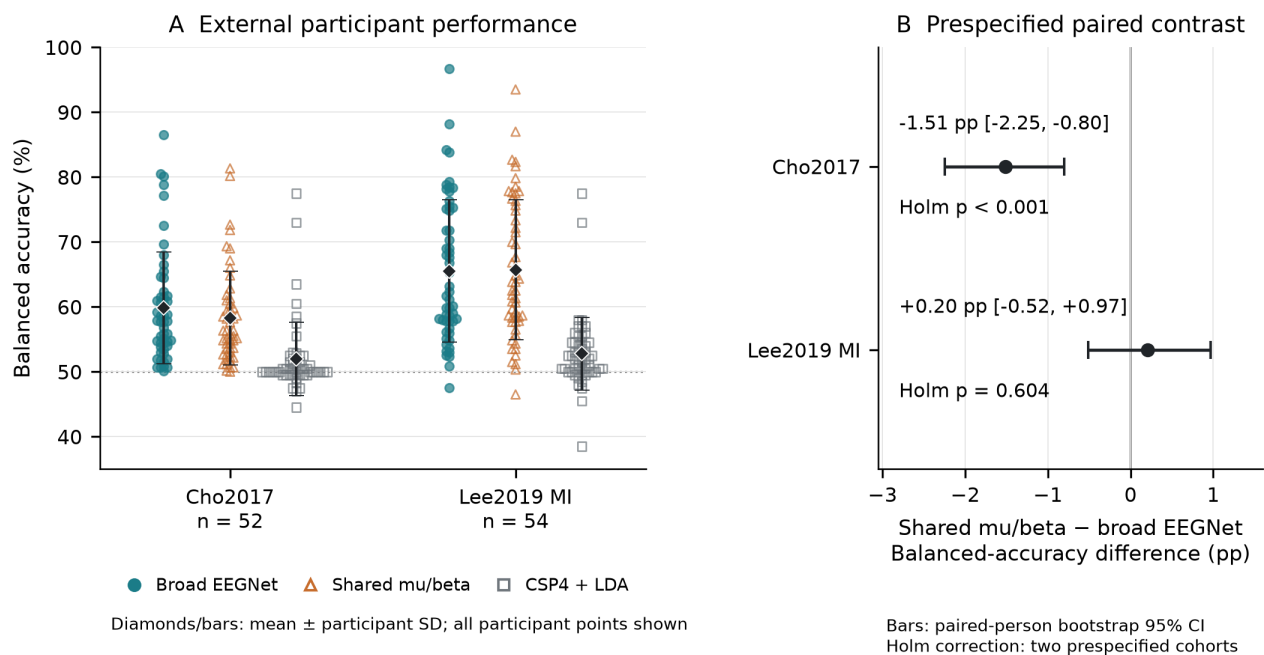


Figure 5. Completed Q15 external evaluation. Cohorts are shown separately, with all person/model scores, equal-person means and person standard deviations, and the frozen paired person-bootstrap 95% intervals. Deep-model scores average three seeds within each person; Lee sessions are pooled before person scoring. Holm p -values belong to the predeclared two-cohort shared-minus-broad family. The common 21-channel, two-second pipeline differs from Q14; no pooled cross-cohort test is shown.

4.7. Descriptive BNCI Sensor Physiology and Decoder Associations (Q16)

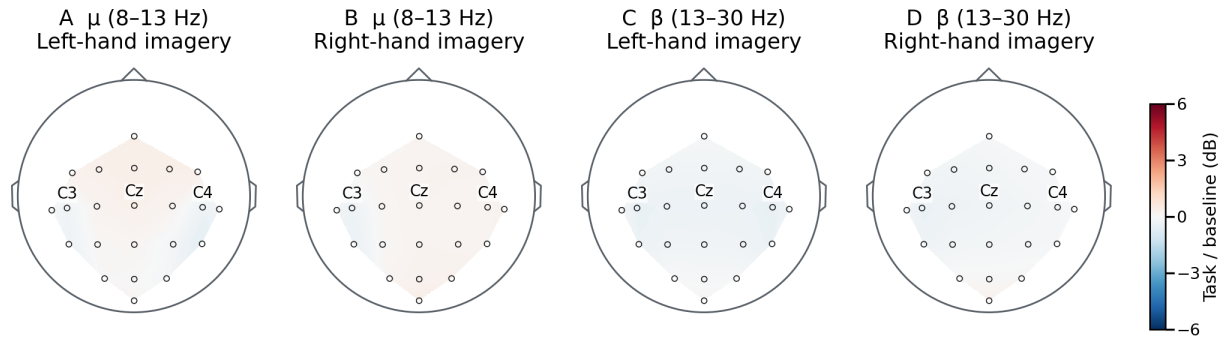
All 18 raw files, 108 labeled runs and 5,184 trials passed the fixed coverage gates. All 228,096 trial-channel-band rows had finite, positive baseline and task power. No trial, session or person was excluded; the 488 provider-flagged trials remained in the primary summaries. Figure 6 shows equal-person scalp means and all nine C3/Cz/C4 profiles. These fixed-estimator task/baseline changes varied by sensor, imagined hand and person.

Table 8. Baseline-relative hand power and signed C3/C4 laterality.

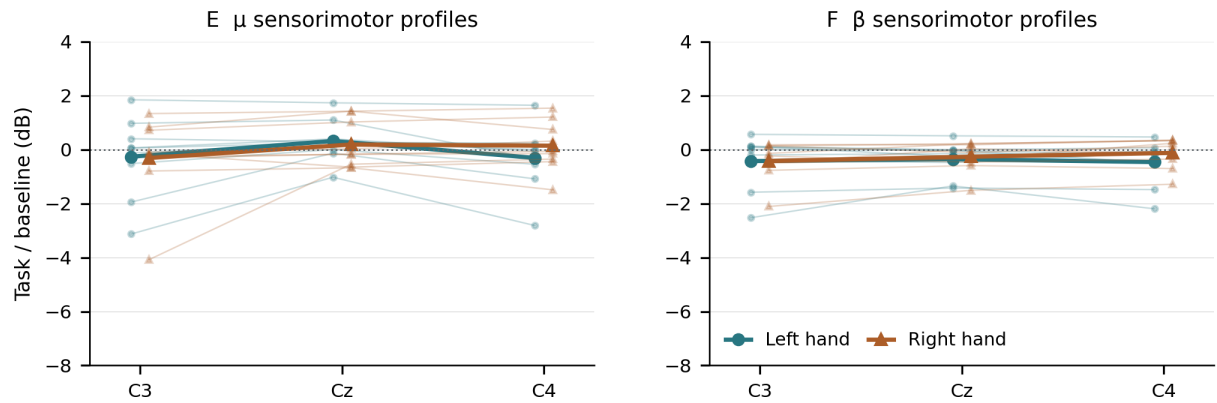
Band	Contra. mean (dB)	Ipsi. mean (dB)	Mean L (dB)	Person L range (dB)	L signs (-/+)
Mu, 8-13 Hz	-0.308	-0.058	-0.250	-1.432 to +0.336	6 / 3
Beta, 13-30 Hz	-0.439	-0.272	-0.168	-0.519 to +0.133	8 / 1

Means give equal weight to nine people, each with equal-session mean trial dB from both hands. Contra. and ipsi. average the relevant C3/C4 hand components; L is contra. minus ipsi. All 2,592 hand trials, including 246 flagged trials, are retained. Negative L indicates a lower contralateral than ipsilateral task/baseline change; it does not establish an absolute decrease in contralateral power. Values are displayed to three decimals; full precision is retained in the supplied data tables. These are descriptive fixed-estimator ratios, not unbiased ERD estimates.

Mu signed laterality averaged -0.250 dB, with six negative and three positive person descriptors; beta averaged -0.168 dB, with eight negative and one positive descriptor (Table 8). Contralateral and ipsilateral group means were both negative in each band, but absolute contralateral changes were not negative for every person. For example, S2 had negative mu laterality (-0.203 dB) while both its contralateral ($+1.491$ dB) and ipsilateral ($+1.694$ dB) means were positive. Separate session, hand and flagged/unflagged cells are supplied in the verified supplementary tables; they are descriptive sensitivity records and do not replace the all-trial population.



Maps: equal-participant means; white areas are outside the 22-sensor convex hull. Template sensor geometry.



Thin lines: all 9 participants; thick lines: participant means. Negative: lower task power. Positive: higher task power. Fixed baseline -1.5 to -0.5 s; imagery $+0.5$ to $+2.5$ s relative to cue. No decoder fitting.

Figure 6. Descriptive native BNCI task/baseline power changes. A-D show equal-person hand-imagery means for mu (8–13 Hz) and beta (13–30 Hz) at all 22 EEG sensors. The schematic standard-1020 geometry has anterior upward and left scalp on viewer left; linear interpolation is restricted to the sensor convex hull. The fixed ± 6 -dB color range clips no sensor or interpolated grid values. E-F show all nine equal-session C3/Cz/C4 profiles (thin lines) and their equal-person means (thick lines). Trial dB uses cue baseline $[-1.5, -0.5]$ s and task $[0.5, 2.5]$ s, with one versus three overlapping Welch segments. Negative values mean lower task power under this estimator, whose unequal window precision can bias log ratios. All trials and artifact flags are retained. Template topography does not localize cortical sources or attribute decoder decisions.

The designated broad-EEGNet descriptive associations between signed laterality and saved binary LOSO balanced accuracy were $\rho = -0.150$ for mu and $+0.600$ for beta. Shared-input coefficients were -0.083 and $+0.483$, and CSP4+LDA coefficients were $+0.067$ and $+0.317$, respectively (Table S4; Figure S3). All six retain the same nine people and have no p-values. A positive beta coefficient means higher balanced accuracy accompanied a larger, less-negative signed descriptor; it does not support a benefit from stronger relative contralateral suppression.

5. Discussion

5.1. Training-duration Sensitivity in a Small Development Benchmark

The most conspicuous internal changes arose from training duration. At two epochs, S3 remained near chance with raw or source-normalized inputs; at 16 epochs, both inputs performed substantially better. Rank aggregation subsequently changed the source-selected schedule and recovered performance for S3 and S8. The matched-runtime control retained a large mean difference between fixed training and the historical mean-loss schedule. These observations make selected duration, source-validation trajectories, per-class recall, and prediction concentration necessary diagnostic outputs when a cross-person neural model appears to collapse.

The evidence supports a diagnostic claim rather than a general prescription. Rank aggregation removes within-fold loss scale but discards magnitude information, and equal fold weighting is itself a choice. The internal candidate was developed after inspecting this benchmark. Moreover, the matched-runtime fixed-duration mean improved while the median person worsened. The external experiments did not compare rank versus mean-loss selection, so they cannot establish independent transfer of the internal rank-selection gain. A prospective comparison must freeze candidate selection rules before new-person outcomes are seen and match runtime, available source examples, and update budgets.

5.2. Complete Spectral Pipelines Have Cohort-dependent Outcomes

The completed external evaluations change the interpretation from an uncertain positive signal to a clear limit on generality. Shared μ/β input gave no internal advantage, a small uncertain positive estimate on PhysioNet, a corrected negative contrast on Cho2017, and an uncertain near-zero contrast on Lee2019. The adverse Cho2017 result is evidence against a consistent advantage of this fixed shared-input pipeline under the evaluated source-only adapter. It does not demonstrate that μ/β rhythms are uninformative, that all parameter sharing is harmful, or that another band definition would succeed.

Class recalls clarify what the cohort means obscure. On Cho2017, the shared-input pipeline had lower mean left-hand recall while increasing right-hand recall; on Lee2019, its corresponding recall changes partly offset. These are descriptive effects of the frozen model, not a post hoc label correction or a physiological mechanism. They are compatible with several unisolated causes, including differences in spatial montage, acquisition reference, exported amplitude scale, temporal alignment, and the constraints imposed by averaging band logits. No intervention in this study identifies which cause produced the adverse contrast.

Parameter economy also differs from compute economy. The shared network has one parameter set but evaluates two band views and shares their BatchNorm statistics. The independent-branch controls change capacity and normalization behavior, while the broadband capacity comparator has a small disclosed parameter mismatch. The source-fitted CSP model transfers poorly and asymmetrically in Q15. Exceeding that contextual comparator cannot be used to claim practical or clinical reliability. None of these experiments is a comparison against contemporary target-adaptation or foundation-model methods under a common budget, and no state-of-the-art claim is made.

The external duration mismatch is part of this interpretation: broad/shared fits used 18/17 epochs in Q14 and 14/19 in Q15. Their paired effects include the selected schedules as well as input views and shared normalization. Neither evaluation identifies the causal contribution of parameter sharing alone.

5.3. Person-level Evidence and Descriptive Signal Characterization

Repeated seeds reduce sensitivity to one stochastic fit but do not add participants. Lee2019 sessions and source-identity subsets are repeated measurements within a person. We consequently use person-level paired effects, retain all declared people, and show the directions of individual differences. The bootstrap intervals condition on the frozen source models; they exclude uncertainty from choosing another nine-person source cohort, another acquisition adapter, or another research-development history. Overlapping outer training sets also complicate a simple independent-person interpretation of the nine-person internal study.

Small-sample procedures can disagree when a few recoveries drive the mean. The rank-selection sign calculation originally retained exact ties, and its conventional tie-excluding sensitivity differs from both the archived calculation and the paired t-test. Reporting this discrepancy avoids giving an erroneous test label scientific authority. The later internal families retain their corrected null findings. Q15 has its own frozen two-cohort family: its Cho2017 result survives that correction, whereas Lee2019 does not establish either benefit or equivalence. PhysioNet uses a different archived primary test and an amended protocol, so its p-value is not silently absorbed into the newer family.

The measured BNCI signals add a sensor-level description without explaining the decoder contrasts. Mean signed hand laterality was negative in both bands: six of nine people had negative mu laterality and eight had negative beta laterality; the remaining signs were positive. Absolute components matter: a negative descriptor can coexist with increased power on both sides, as observed in S2. The broad beta correlation was positive, so higher balanced accuracy coincided with less-negative relative laterality rather than stronger suppression. None of the six small-sample associations supports a subgroup threshold or model selection.

A lateralized group-average scalp pattern or a correlation with LOSO balanced accuracy does not establish a cortical source, exclude ocular or muscle contributions, or show that EEGNet used the depicted signal. The physiology was specified after decoder outcomes were known and concerns BNCI only. It therefore cannot explain the adverse Cho contrast or validate an external physiological mechanism.

5.4. Limitations

The same nine BNCI people informed successive development conditions. Per-fit target exclusion does not remove outcome-informed candidate design or dependence through overlapping LOSO training sets. The internal rank-selection gain was concentrated in two people, and external stages did not test that selection-rule contrast. Several controls differ in runtime, training examples, update exposure, sensor width or approximate capacity. No common-budget comparison against contemporary target-adaptation or foundation-model methods was performed.

Q14 and Q15 use distinct montages, windows, references, filter scopes and source checkpoints. Q14's metadata amendment followed partial execution. The frozen transfer experiment's repository freezes precede its fitting or inference stages but follow metadata inspection and are not independently registered preregistration. Cross-provider participant identities are unverified. External bootstrap intervals condition on one frozen source cohort and do not include source-sampling, adapter or development-history uncertainty.

The external source-duration schedules were selected from four fixed BNCI validation groups with unequal participant counts (2/2/2/3) and equal fold weighting. This grouping was frozen before target inference but represents one arbitrary source-validation design. Grouping

sensitivity was not evaluated prospectively.

Q15 used a common operational tensor and a committed adapter to make transfer executable across three providers. A common tensor does not establish a common physical measurement. The native export voltage calibration, original Cho acquisition reference, and hardware cue latency were not independently verified. Cho channel order is an explicitly reviewed mapping from the publication montage because the MATLAB payload lacks channel-name strings. Lee event indexing retains a one-native-sample convention discrepancy. These limitations are part of the method’s scope, rather than defects that can be erased by a passing numerical validator.

The distinction between Q14 and Q15 matters here. The earlier PhysioNet run required a sampling-rate amendment after partial execution, and its later portability validator rechecked existing outputs without replaying inference. Q15 separately froze the preprocessing contract before source fitting and the inference contract before external predictions; its completed scientific validator reports independent raw-to-prediction replay. The manuscript’s own checks only recalculate saved outputs and verify their hash bindings. Keeping these levels separate lets a reader assess reproducibility without mistaking a launcher, a saved status flag, or an API response for scientific validation.

All models were evaluated offline with zero-phase filtering. No online latency, prospective assistive-control performance, clinical population, or patient benefit was tested. The two-second Q15 crop also excludes part of each imagery interval and does not reproduce each provider’s original benchmark. A useful next experiment would preregister a fully calibrated adapter and compare frozen source-only models on another cohort, with per-class and per-person outcomes specified before access. Such a study would test the boundary revealed here; the present results do not justify retuning on the held-out cohorts and calling the resulting performance external confirmation.

The physiological analysis uses a one-segment baseline and a three-segment task estimator; unequal log-power variance can bias their dB difference. Its signed descriptor measures relative contralateral-ipsilateral change and can be negative through ipsilateral increases without absolute contralateral suppression. The descriptor and binary decoder also use different task windows. Nine descriptive person records and overlapping LOSO fits cannot establish a subgroup biomarker or mechanistic model attribution. Baseline-relative ratios cancel only a constant multiplicative gain and do not correct reference, timing, spatial mapping or artifacts. External native voltage calibration remains unknown, and external physiology is unrun.

6. Conclusion

Source-only training-duration choices produced large but concentrated gains in the nine-person BNCI development benchmark. The evaluated shared-input pipelines showed no consistent external advantage: compared with broad EEGNet, balanced accuracy was 1.513 pp lower on Cho2017, whereas the Lee2019 contrast remained compatible with a small effect in either direction. These complete-pipeline comparisons include unequal source-selected durations and do not isolate parameter sharing. The post-outcome BNCI signal analysis found negative mean signed hand laterality in mu and beta with substantial participant variation; it does not establish a decoder mechanism or external physiological replication. Unknown acquisition calibration and offline evaluation limit deployment conclusions.

Data and code availability

Public BNCI2014_001, PhysioNet, Cho2017 and Lee2019 recordings are available from the original providers [1,2,5,6,7,8,9,10], subject to their licenses and terms of use. PhysioNet platform publications are cited separately [27,28]. Analysis code, saved predictions, participant-level summaries, parameter specifications and numerical validation reports are available at <https://github.com/jackzhu119/cross-subject-mi-eeg>. The supplementary reproducibility documentation identifies the exact source and result revisions. Raw EEG is not redistributed in the manuscript bundle.

Ethics statement

This study involved secondary analysis of public EEG recordings and recruited no new participants. The original studies report their collection ethics and participant consent. The author confirms that neither ethics approval nor an exemption was required for this secondary analysis.

Funding

This research received no funding.

Competing interests

The author declares no competing interests.

Acknowledgements

AI assistance disclosure

OpenAI Codex assisted source and literature checks, analysis-code development, figure preparation, and manuscript drafting and language revision. The author is responsible for reviewing the final manuscript and for its scientific content.

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